

Mucocutaneous and Systemic Manifestations of Congenital Syphilis

Initially thin and mucoid, nasal discharge in congenital syphilis—teeming with spirochetes—can become purulent or bloody. The nasal septum may eventually perforate. If untreated, destruction of the nasal cartilage leads to flattening of the nasal bridge, resulting in the characteristic **saddle nose** deformity.

The skin of the syphilitic neonate is often dry and wrinkled. In newborns with fair skin, it may exhibit a **café-au-lait hue**. Mucocutaneous manifestations occur in approximately half of patients under 6 months of age and resemble lesions of secondary syphilis. These typically present as copper- to red-colored macules and papules, with or without scaling, predominantly affecting the **palms, soles, and diaper area**. **Condylomata lata** and **mucous patches** may also be present. Annular or corymbiform eruptions have been reported.

Bullae, 1 to 5 cm in diameter, are rare but highly infectious and may be generalized or, more commonly, localized to the extremities—especially the palms and soles. Known as **syphilitic pemphigus**, the presence of blisters indicates severe disease. Pustules may develop on the fingers, toes, and around the angles of the mouth. Ulcerations near the mouth, nose, or anus can heal into **rhagades (Parrot's lines)**—depressed linear scars that persist into adulthood and radiate from orifices like the spokes of a wheel.

Lymphadenopathy and Skeletal Involvement

Lymphadenopathy occurs in about half of cases, though it may not be prominent. The presence of firm, movable, nontender **epitrochlear lymph nodes** is particularly characteristic of prenatal syphilis.

Bone disease is the most common early manifestation, found in up to **97% of autopsied infants under 6 months**. Although often asymptomatic, it can lead to significant morbidity. The hallmark lesion is **osteochondritis**, identifiable on radiographs by a **sawtooth appearance in the metaphysis**. A classic finding is the **Wimberger sign** (also known as the "catbite" sign)—a

radiolucent defect at the medial aspect of the proximal tibial metaphysis—seen in about 20% of early cases.

Longitudinal lines of rarefaction resembling "**celery sticks**" may extend from the metaphysis into the diaphysis. Pain from osteochondritis or epiphysitis worsens with movement, leading the infant to immobilize the affected limb—a clinical sign known as **pseudoparalysis of Parrot**.

Painful **periostitis**, frequently observed in the second half of the first year, can result in blunting of the scapular spines and anterior tibial margins later in life. The deposition of multiple layers of new bone gives rise to the "**onion-peel periosteum**" appearance on imaging. **Dactylitis** due to osteitis of the fingers is uncommon but may occur within the first two years of life.

Central Nervous System Involvement

Spirochetes are detected in the **central nervous system (CNS)** in **40% to 60%** of infants with congenital syphilis, highlighting the importance of neurologic evaluation and appropriate treatment to prevent long-term sequelae.